

BLOOD CANCER DETECTION USING HYBRID ENSEMBLE DEEP LEARNING

ABSTRACT

Blood cancer, particularly leukemia, affects the production and function of white blood cells. Early diagnosis is essential for effective treatment and improved patient outcomes. Manual examination of blood smear images is time-consuming and dependent on expert knowledge. This project presents an automated blood cancer detection system using deep learning and ensemble learning techniques. Three deep learning models, namely ResNet50, DenseNet121, and InceptionV3, are used to classify blood cell images into Leukemia and Normal classes. To improve prediction reliability, Hard Voting, Soft Voting, and Learnable Ensemble techniques are employed. Class imbalance is handled using class weights during training. Explainable Artificial Intelligence (XAI) is incorporated through Grad-CAM to visualize the regions influencing the model's decision. Experimental results demonstrate that the proposed ensemble approach improves classification performance while providing interpretable predictions.

INTRODUCTION

Blood cancer is a disease that affects the formation and function of blood cells. Leukemia is one of the most common blood cancers and originates in the bone marrow. Early detection plays a crucial role in successful treatment.

Traditional diagnosis relies on microscopic examination by medical experts, which can be time-consuming and prone to human error. Deep learning techniques have shown significant success in medical image analysis and can assist doctors in faster and more accurate diagnosis.

The objective of this project is to develop an automated blood cancer detection system using deep learning models and ensemble learning techniques while also providing visual explanations for predictions.

Objectives

- Detect leukemia from blood smear images.
- Improve prediction accuracy using ensemble learning.
- Handle class imbalance effectively.
- Provide explainable predictions using Grad-CAM.
- Develop a reliable single-image prediction system.

DATASET DESCRIPTION

The C-NMC Leukemia Dataset was used for this project. The dataset contains microscopic blood smear images belonging to two categories: **Leukemia** and **Normal**. The images were collected from real clinical samples and provide a suitable benchmark for evaluating automated leukemia detection systems.

The dataset contains a total of 10,661 images. Among these, 8,528 images were used for training and 2,133 images were used for validation. The dataset was divided using an 80:20 ratio to ensure proper model evaluation while maintaining sufficient training samples.

The class mapping used in this project is:

- **Leukemia** → 0
- **Normal** → 1

Due to unequal class distribution, class weights were calculated and applied during training. The calculated class weights were:

- **Leukemia: 0.7330**
- **Normal: 1.5729**

These weights ensure balanced learning and prevent the model from favoring the majority class.

```
Found 8528 validated image filenames belonging to 2 classes.  
Found 2133 validated image filenames belonging to 2 classes.  
Class Mapping: {'leukemia': 0, 'normal': 1}
```

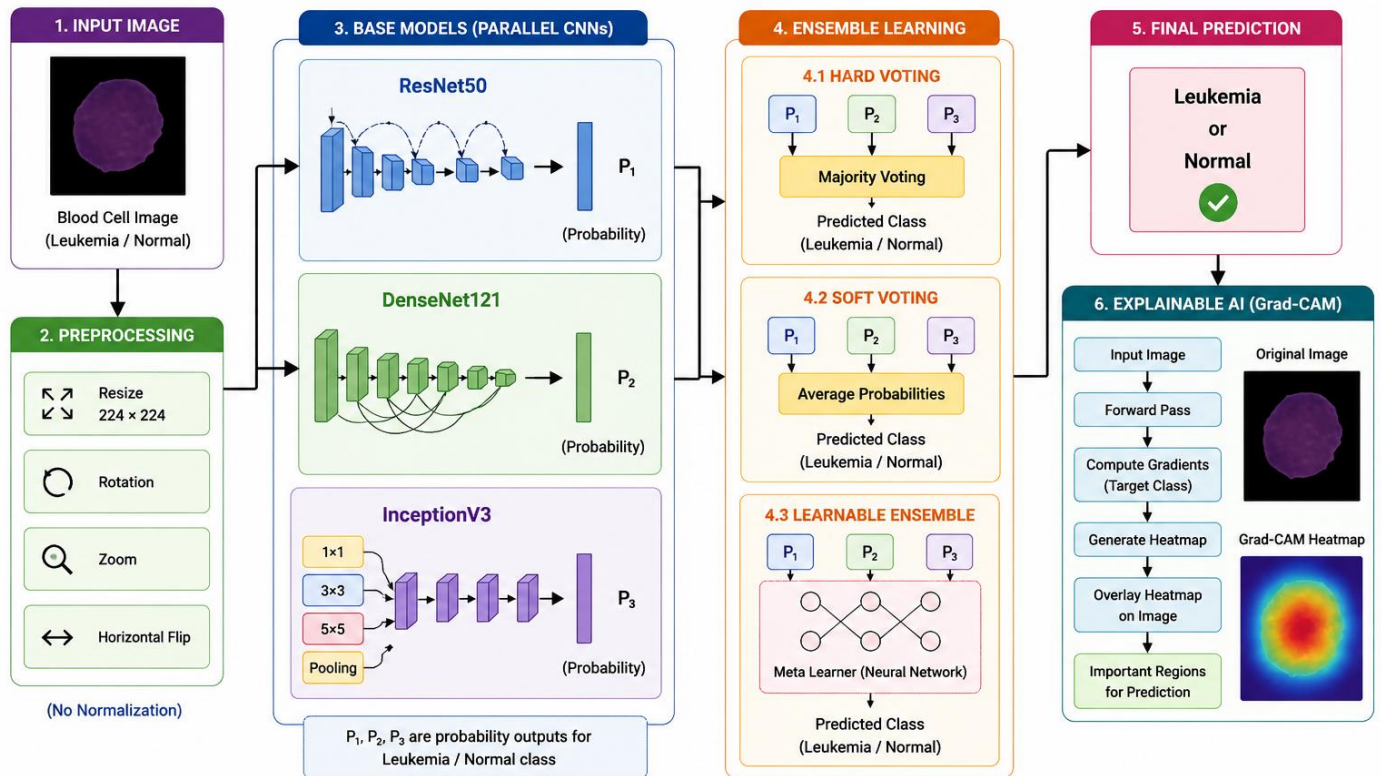
CLASS WEIGHTS:

```
{0: np.float64(0.7330238954787691), 1: np.float64(1.5728513463666545)}
```

SYSTEM ARCHITECTURE

ARCHITECTURE DIAGRAM

Blood Cancer (Leukemia) Detection using Hybrid Ensemble Deep Learning with Explainable AI



METHODOLOGY

Data Preprocessing

Data preprocessing is an essential step that improves the quality and diversity of training samples. All images were resized to 224×224 pixels. Data augmentation techniques including rotation, zooming, and horizontal flipping were applied to increase dataset diversity and reduce overfitting.

Handling Class Imbalance

Medical datasets frequently suffer from class imbalance. Training directly on imbalanced data may cause models to favor the majority class. To overcome this issue, class weights were computed using the inverse frequency of each class and applied during model training.

Deep Learning Models

Three deep learning models were employed for feature extraction and classification.

ResNet50 utilizes residual connections that help overcome the vanishing gradient problem and enable effective training of deeper networks.

DenseNet121 establishes dense connectivity between layers, improving feature reuse and information flow throughout the network.

InceptionV3 uses parallel convolution operations with multiple kernel sizes, allowing the network to capture fine and large-scale features simultaneously.

Ensemble Learning

Three ensemble techniques were implemented:

Hard Voting: Majority voting among model predictions.

Soft Voting: Average probability generated by all models.

Learnable Ensemble: A meta-neural network trained on prediction probabilities generated by ResNet50, DenseNet121, and InceptionV3.

Explainable Artificial Intelligence

Grad-CAM was used to provide visual explanations for model predictions. This technique computes gradients of the target class with respect to the final convolutional feature maps and generates heatmaps highlighting important image regions.

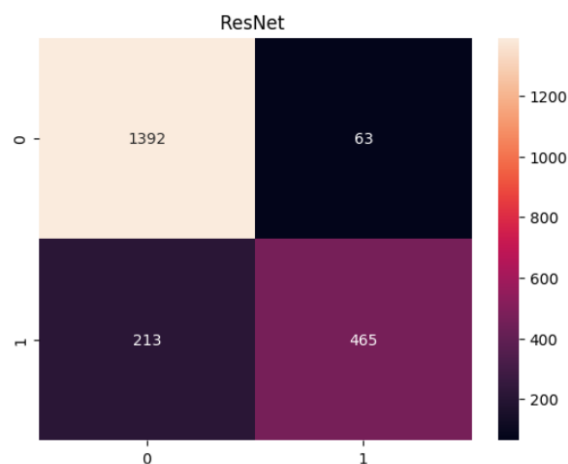
RESULTS

ResNet50 Performance

The ResNet50 model achieved an accuracy of 87.06% on the validation dataset. The confusion matrix indicates that the model correctly classified 1392 leukemia samples and 465 normal samples.

67/67 — 8s 115ms/step

ResNet	Accuracy: 0.870604781997187				
	precision	recall	f1-score	support	
0	0.87	0.96	0.91	1455	
1	0.88	0.69	0.77	678	
accuracy			0.87	2133	
macro avg	0.87	0.82	0.84	2133	
weighted avg	0.87	0.87	0.87	2133	

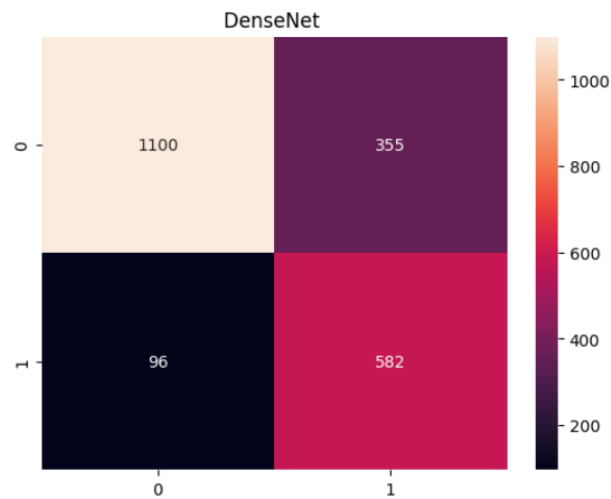


DenseNet121 Performance

DenseNet121 achieved an accuracy of 78.86%. Although it demonstrated strong recall for the normal class, it produced more false positive predictions than the other models.

67/67 ————— 5s 74ms/step

DenseNet	Accuracy: 0.7885607126113455				
	precision	recall	f1-score	support	
0	0.92	0.76	0.83	1455	
1	0.62	0.86	0.72	678	
accuracy			0.79	2133	
macro avg	0.77	0.81	0.78	2133	
weighted avg	0.82	0.79	0.80	2133	

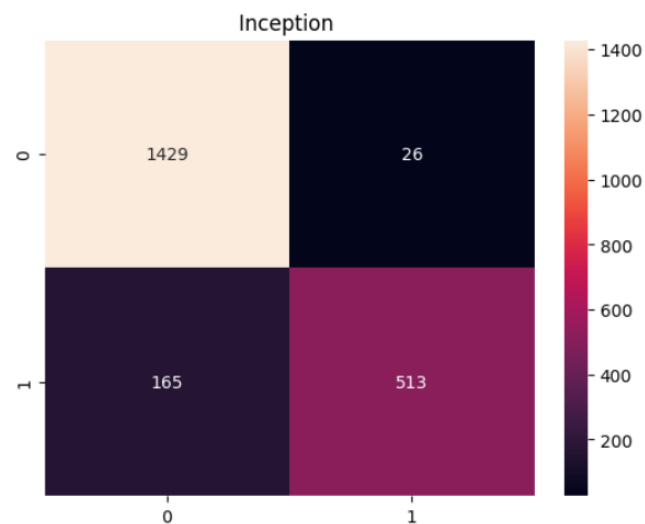


InceptionV3 Performance

InceptionV3 achieved the highest individual model accuracy of 91.05%. Its multi-scale feature extraction capability enabled superior classification performance.

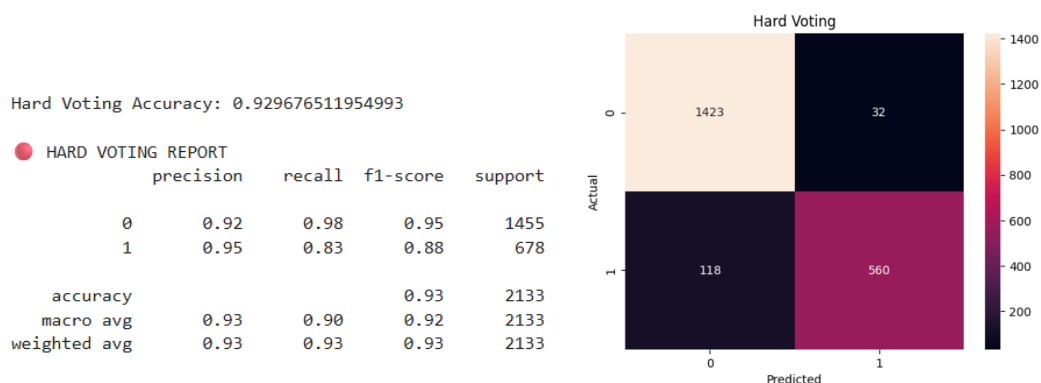
67/67 ————— 17s 163ms/step

Inception	Accuracy: 0.9104547585560244				
	precision	recall	f1-score	support	
0	0.90	0.98	0.94	1455	
1	0.95	0.76	0.84	678	
accuracy			0.91	2133	
macro avg	0.92	0.87	0.89	2133	
weighted avg	0.91	0.91	0.91	2133	



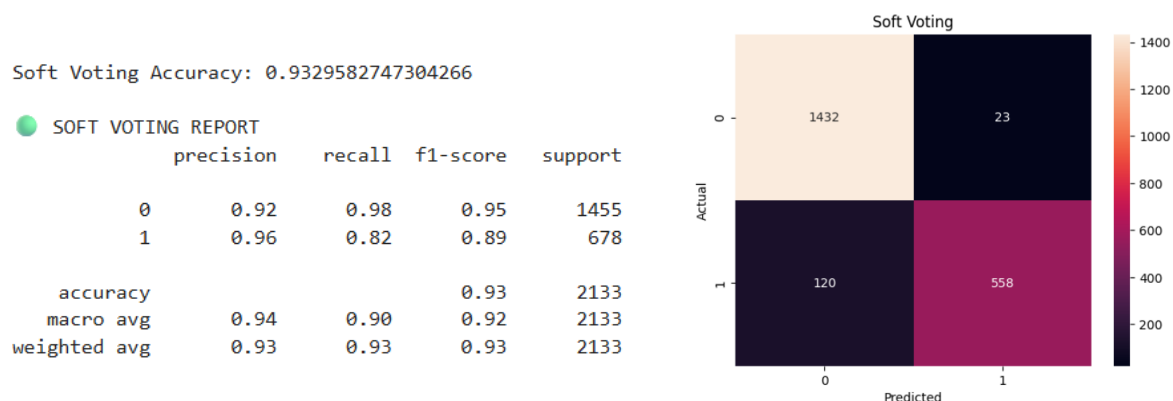
Hard Voting Ensemble Performance

The Hard Voting Ensemble achieved an accuracy of 92.97%, improving performance by combining predictions from all three models.



Soft Voting Ensemble Performance

The Soft Voting Ensemble achieved the highest overall accuracy of 93.30%. Averaging prediction probabilities produced more stable and reliable results.



Learnable Ensemble Performance

The Learnable Ensemble achieved an accuracy of 92.92%. The meta-neural network successfully learned optimal weights for combining model predictions.

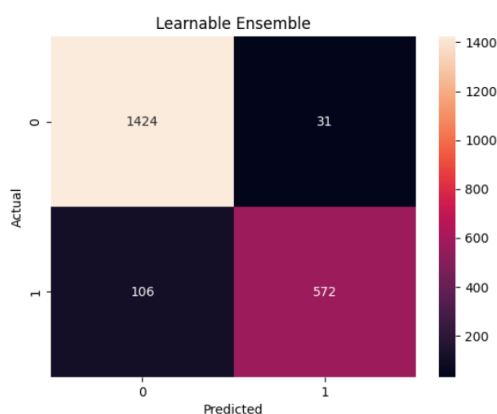
67/67 — 0s 4ms/step

WARNING:absl:You are saving your model as an HDF5 file via `model.save()` which is considered legacy. We recommend using instead the native Keras API: `model.save('my\_model.keras')`.

Learnable Ensemble Accuracy: 0.9292076887013596

● LEARNABLE ENSEMBLE REPORT

	precision	recall	f1-score	support
0.0	0.93	0.97	0.95	1455
1.0	0.93	0.84	0.88	678
accuracy			0.93	2133
macro avg	0.93	0.90	0.92	2133
weighted avg	0.93	0.93	0.93	2133



Single Image Prediction

A single blood cell image was provided to the system for testing. Predictions generated by ResNet50, DenseNet121, and InceptionV3 were combined using the ensemble methods. All ensemble techniques classified the sample as Leukemia.

1/1 ————— 0s 60ms/step
1/1 ————— 0s 60ms/step
1/1 ————— 0s 61ms/step
1/1 ————— 0s 44ms/step

MODEL OUTPUTS

ResNet: 0.0056088716
DenseNet: 0.99059266
Inception: 0.17108288

ENSEMBLE RESULTS

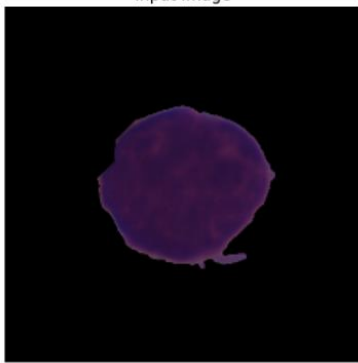
Hard Voting: 0
Soft Voting: 0
Learnable Ensemble: 0

✓ FINAL PREDICTION: leukemia

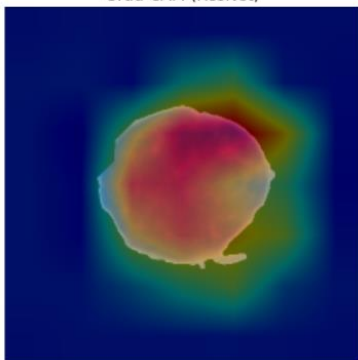
Explainable AI Analysis

Grad-CAM heatmaps were generated to visualize the regions influencing model predictions. The generated heatmap highlights the cellular region, indicating that the model focuses on medically relevant features rather than background information.

Input Image



Grad-CAM (ResNet)



CONCLUSION

This project successfully developed an automated blood cancer detection system using hybrid ensemble deep learning and explainable artificial intelligence techniques. Three convolutional neural network architectures, namely ResNet50, DenseNet121, and InceptionV3, were trained on the C-NMC Leukemia Dataset. Class imbalance was effectively addressed through class weighting techniques. Ensemble learning methods including Hard Voting, Soft Voting, and Learnable Ensemble were implemented to improve classification reliability.

Among the individual models, InceptionV3 achieved the highest accuracy of 91.05%. The Soft Voting Ensemble produced the best overall performance with an accuracy of 93.30%. Furthermore, Grad-CAM provided visual explanations for model decisions, improving transparency and interpretability. The obtained results demonstrate that ensemble deep learning combined with explainable AI can serve as an effective tool for assisting healthcare professionals in leukemia diagnosis and decision-making.